

7. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

surface area of cube = length $\times$ width $\times$ number of sides

volume of cube = length $\times$ width $\times$ height

- (a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm ²)	Volume of cube (cm ³)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150		6:5
7	104		343	6:7
10	610	600	1000	

- (i) **Complete** the table. [3]

- (ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

.....

.....

.....

(b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

.....

.....

.....

.....

END OF PAPER

Examiner
only

8

Examiner
only

For continuation only.

Examiner
only

For continuation only.

BLANK PAGE

BLANK PAGE

THE PERIODIC TABLE

Group

1 2

3

4

5

6

7

0

1 H Hydrogen 1																	
7 Li Lithium 3	9 Be Beryllium 4																
23 Na Sodium 11	24 Mg Magnesium 12																
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number